

- Aim:
- To determine the COD of given water sample by open reflux method.
- chemicals:
- Potassium dichromate, sulphuric acid, ferrous ammonium sulphate, silver sulphate, mercury sulphate, ferroin indicator.
- Theory:
- Chemical oxygen Demand (COD) test determines the oxygen requirement equivalent of organic matter that is susceptible to oxidation with the help of a strong chemical oxidant. It is an important, rapidly measured parameter as a mean of measuring organic strengths for streams and polluted water bodies. The test can be related empirically to BOD, organic carbon or organic matter in samples from a specific source taking into account its limitations. The test is useful in studying performance evaluation of wastewater treatment plants and monitoring relatively polluted water bodies. COD determination has advantages over BOD determination. COD results can be obtained in 3-4 hours as compared to 3-5 days required for BOD determination. Further, the test is relatively easy, precise, and is unaffected by interferences as in BOD test. The intrinsic limitation of the test lies in its inability to differentiate between the biologically oxidizable and biologically inert material and to find out the system rate constant of aerobic biological stabilization.
- The open reflux method is suitable over a wide range of wastes with a large sample size. The dichromate reflux method is preferred over procedures using other oxidants (e.g. potassium permanganate) because of its superior oxidizing ability, applicability to a wide range of samples.

° Observations:

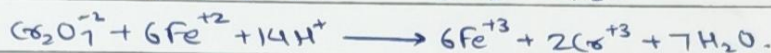
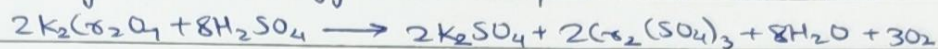
Sr.No.	Sample Name	Volume of sample	Burette Reading		Volume of FAs used
			Initial	Final	
1	Distilled	25mL	0	14.5	14.5
2	Distilled	25mL	0	14.7	14.7
3	Waste	25mL	0	8.5	8.5
4	Waste	25mL	8.5	16.5	8

° Calculations:

$$a = \frac{14.5 + 14.7}{2} = 14.6 \quad b = \frac{8.5 + 8}{2} = 8.25$$

$$\therefore \text{COD} = \frac{(a-b) \times N \times 8000}{V} = \frac{(14.6 - 8.25) \times 0.25 \times 8000}{25} = 508 \text{ mg/L}$$

and ease of manipulation. Oxidation of most organic compounds is up to 95-100% of the theoretical value. The organic matter gets oxidized completely by potassium dichromate ($K_2Cr_2O_7$) with silver sulphate as catalyst in presence of concentrated H_2SO_4 to produce CO_2 and H_2O . The excess $K_2Cr_2O_7$ remaining after the reaction is titrated with ferrous ammonium sulphate ($[Fe(NH_4)_2(SO_4)_2]$). The dichromate consumed gives the oxygen (O_2) required for the oxidation.



Procedure:

- Take 25ml of water sample into a conical flask.
- Add 10ml of $K_2Cr_2O_7 / HgSO_4$ solution to the flask.
- Add 10ml of H_2SO_4 / Ag_2SO_4 solution through sides of the flask.
- Transfer the solution to a round bottom flask. Add 10ml of distilled water to rinse the conical flask and transfer to RBF.
- Reflux the sample for 1.5 hours; meanwhile fill burette with FAS.
- Transfer the solution from RBF to conical flask completely. Add 10ml of distilled water to rinse the RBF and transfer to conical flask.
- Add about 5-6 drops of ferroin indicator. The solution will be greenish blue by now.
- Titrate it against FAS. The endpoint is appearance of reddish brown color.

Result:

- The chemical oxygen Demand of given water sample is 508 mg/L.